AIRLINE FLIGHT
OPERATIONS DASHBOARD

BUSINESS REQUIREMENTS DOCUMENT

Real-time Flight-Schedule Analysis
using the **AviationStack API**

Real-time Data	Data Analysis & Insights	Interactive Dashboards	Operational Visibility

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Data Source
AviationStack API (Free Tier)

Tools Used
Python (requests, pandas),
HTML/CSS, Power BI

Document Version
1.0

“Turning real-time flight data into actionable insights for smarter operational decisions.”

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1. Introduction & Background

This Business Requirements Document (BRD) was produced as part of a self-directed portfolio project undertaken while preparing for a Business Analyst internship. Rather than working from a static, pre-cleaned dataset, the project was designed to mirror a realistic analyst task: connect to a live external data source, understand what it can and cannot support, and turn it into a decision-ready report for a non-technical stakeholder.

The subject chosen — airline flight-schedule data — was selected because it produces rich, multi-dimensional records (airline, route, timing, geography) from a single free API, making it well suited to demonstrate a full analysis lifecycle: extraction, cleaning, aggregation, visualization, and documentation.

This document is intended to be read alongside the accompanying Power BI dashboard and the custom-built HTML/CSS dashboard, both of which were produced from the same underlying dataset described here.

2. Executive Summary

This document outlines the business context, objectives, methodology, and findings of a self-directed project undertaken to demonstrate business analysis and data-handling capability using a live, third-party data source. The project pulls real-time flight-schedule data through the AviationStack API, processes and analyzes it using Python, and presents the results through both a Power BI dashboard and a custom HTML/CSS dashboard.

The purpose of this exercise is to simulate a realistic business analyst workflow: sourcing data from an external system, understanding its structure and limitations, extracting operationally relevant insights, and communicating those insights clearly to a non-technical audience.

3. Business Objective

Airlines and airport operations teams rely on up-to-date flight information to plan resources, communicate with passengers, and identify operational bottlenecks. This project simulates a scenario in which an operations or planning team requests a quick, visual snapshot of current scheduled flight activity across airlines and departure airports, in order to answer questions such as:

- Which airlines have the highest volume of scheduled flights in the current data pull?
- Which departure airports are busiest, and could this indicate scheduling or resourcing pressure?
- What is the geographic spread of departure activity?
- Is the available data sufficient to assess on-time performance, or are there gaps that a stakeholder should be aware of?

4. Stakeholders

In a live deployment of this kind of dashboard, the following stakeholder groups would be the primary audience. For this portfolio version, these are simulated roles used to frame the requirements and design decisions:

Stakeholder	Interest	Key Question
Operations Planning	Resource allocation across airports	Where is departure volume concentrated?
Network Planning	Route performance	Which routes carry the most traffic?
Customer Experience	Passenger communication timing	When are most departures scheduled?
Executive Sponsor	High-level KPIs	Total flights, airlines, and geographic reach

5. Scope

In scope

- Extracting real-time flight data via a public API (100 records, free-tier limit).
- Cleaning and structuring the data using Python (pandas).
- Producing summary statistics: flight counts by airline, by departure airport, and by status.
- Building two visual outputs: a Power BI dashboard and a custom HTML/CSS dashboard.

Out of scope

- Historical trend analysis (the free-tier API does not provide historical records).
- Verified on-time performance / delay analysis (see Limitations, Section 6).
- Live/auto-refreshing data (this is a single point-in-time snapshot).

6. Assumptions & Constraints

Assumptions

- The 100-record sample returned by the API at the time of the pull is treated as representative of "a typical snapshot," not of global flight volume.
- Airport country/continent classifications were mapped manually against the 18 distinct departure airports observed, based on publicly known airport locations.

Constraints

- Free-tier API access limits requests to 100 per month, which ruled out repeated pulls for trend analysis within this project's timeframe.
- No budget was allocated for a paid API tier or paid geocoding service; where arrival-airport coordinates could not be confidently matched against a public airport database, those routes were excluded from map visuals rather than estimated.

7. Methodology

7.1 Data extraction

A Python script authenticates against the AviationStack REST API and requests up to 100 real-time flight records. The raw JSON response is parsed and flattened into a tabular structure containing flight date, status, airline, flight number, departure/arrival airports, and scheduled/delay timestamps.

7.2 Data processing

The flattened dataset is loaded into pandas for cleaning and aggregation. Key transformations include grouping by airline, departure airport, country, and continent to produce frequency counts; extracting the hour component from scheduled departure timestamps to build a time-of-day distribution; and checking for missing/null values in delay fields to assess data completeness.

7.3 Geographic matching

Departure airport coordinates were compiled manually for the 18 distinct airports observed. Arrival airport names were cross-referenced against a public airport database (name-based matching) to obtain coordinates for route-mapping; 34 of 39 distinct arrival airports (87%) were successfully matched. The remaining 13% were excluded from route visuals rather than approximated, consistent with the project's no-fabrication principle.

7.4 Visualization

Two dashboards were produced from the same underlying dataset:

- Power BI: bar charts for flights by airline and by departure airport, a status breakdown chart, and summary cards.
- Custom HTML/CSS dashboard: a filterable KPI summary, ranked airline/airport/country lists, a real-geography route map, an hourly departure-time chart, and a status donut — built to practice translating a dashboard concept into functioning front-end code with live client-side filtering.

8. Key Findings

Metric	Value	Notes
Total flights captured	100	Single API pull, free-tier cap
Distinct airlines represented	42	High fragmentation, no single dominant carrier
Distinct departure airports	18	Spread across Asia-Pacific primarily
Top airline (tied)	Xiamen Air / China Express / Aeroflot	6 flights each

Metric	Value	Notes
Busiest departure airport	Tianjin Binhai International	32 flights (32% of total)
Flights with 'scheduled' status	100 (100%)	See limitation below

Observation: Tianjin Binhai International and Wenzhou together account for 52% of all departures in this snapshot, despite representing only 2 of the 18 airports observed. In a live operational setting, this kind of concentration would be a useful early signal for prioritizing resourcing or contingency planning at those locations.

9. Data Limitations

A core part of business analysis is being transparent about what the data can and cannot support. The following limitations were identified during this project:

- Free-tier snapshot only: All 100 records were captured at a single point in time and carried a status of “scheduled.” None of the flights had yet departed or landed, so arrival delay data was present for only 2 of 100 records — not enough to calculate a reliable on-time performance percentage.
- No historical data: The free AviationStack plan excludes historical flights, which prevents trend analysis over time (e.g., week-over-week delay patterns).
- Sampling bias: Because the API returns whatever flights are active at request time, the dataset is skewed toward regions where it was daytime/peak activity during the pull (in this case, East Asia), rather than a representative global sample.

Recommendation: for a production use case, this would require either a paid API tier with historical access, or a scheduled batch process that pulls and stores data at regular intervals to build a genuine time series.

10. Risks & Mitigations

Risk	Impact if unaddressed	Mitigation
Snapshot mistaken for a trend	Stakeholders over-generalize from one moment in time	Explicit labeling as a snapshot throughout
Fabricated/estimated KPIs	Erodes trust if numbers can't be traced to source	Every metric traced to raw data; gaps documented, not filled
API quota exhaustion	Free tier limited to 100 requests/month	Single, deliberate pull; results cached to CSV
Incorrect route geolocation	Misleading map if airport coordinates	Only reference-database-

Risk	Impact if unaddressed	Mitigation
	are wrong	matched airports are plotted

11. Recommendations

- Schedule automated pulls (e.g., hourly) and store results in a database to enable real trend and delay analysis over time.
- Upgrade to a paid API tier if delay/on-time analysis becomes a business requirement, since the free tier cannot support it.
- Extend the dashboard with drill-through filters (by airline, by airport, by date) once a larger, time-distributed dataset is available.

12. Conclusion

This project demonstrates an end-to-end business analysis workflow: defining a business question, sourcing and understanding a real external data feed, being explicit about its limitations, extracting actionable insights, and presenting them through both a standard BI tool (Power BI) and custom-built visual reporting (HTML/CSS). The emphasis throughout was on accuracy and honesty in reporting — no metric shown in the accompanying dashboards is estimated or fabricated; every figure is traceable back to the raw API data.

Appendix A — Data Dictionary

Fields extracted from the AviationStack API response and used in this analysis:

Field	Type	Description
flight_date	Date	Calendar date of the scheduled flight
flight_status	Text	Status at time of pull (e.g., scheduled, active, landed, cancelled)
airline_name	Text	Operating airline
flight_number	Text	IATA flight number / code
departure_airport	Text	Origin airport name
departure_scheduled	Timestamp	Scheduled departure time (UTC, ISO 8601)

Field	Type	Description
departure_delay	Number (min)	Departure delay in minutes, where reported
arrival_airport	Text	Destination airport name
arrival_scheduled	Timestamp	Scheduled arrival time (UTC, ISO 8601)
arrival_delay	Number (min)	Arrival delay in minutes, where reported (sparse — see Section 9)

Appendix B — Sample Raw Data & Dashboard Preview

First 5 records of the raw dataset (unmodified, as returned by the API):

Status	Airline	Flight #	Departure	Arrival
scheduled	Ryanair	FR	Copernicus Airport	Bourgas
scheduled	Shanghai Airlines	FM9553	Wenzhou	Xianyang
scheduled	China United Airlines	KN3025	Wenzhou	Xianyang
scheduled	China Eastern Airlines	MU8557	Wenzhou	Xianyang
scheduled	China United Airlines	KN5289	Wenzhou	Changchun

Custom HTML/CSS Dashboard — Overview

Revision History

Version	Date	Change
1.0	July 2026	Initial BRD: objectives, scope, methodology, findings, limitations, recommendations
2.0	July 2026	Added Introduction, Stakeholders, Assumptions & Constraints, Risks & Mitigations, Data Dictionary, Sample Data & Dashboard Preview appendices; added page numbering and expanded Key Findings with country/continent/hour/route analysis